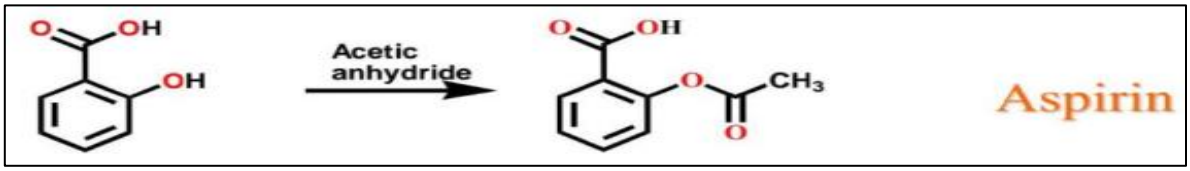
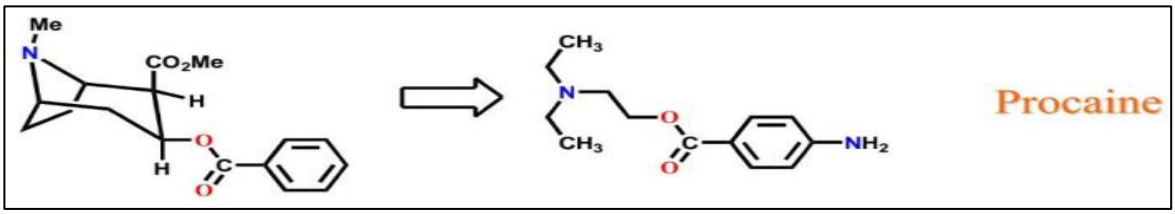
✓ Willow tree - salicylic acid



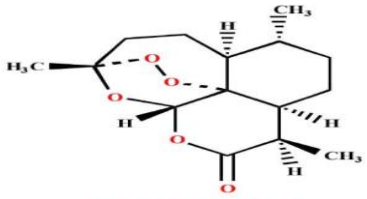
✓ Coca bush – cocaine



✓ **Plants and ancient records**

- **Artemisinin**

  lead compound for anti-malaria

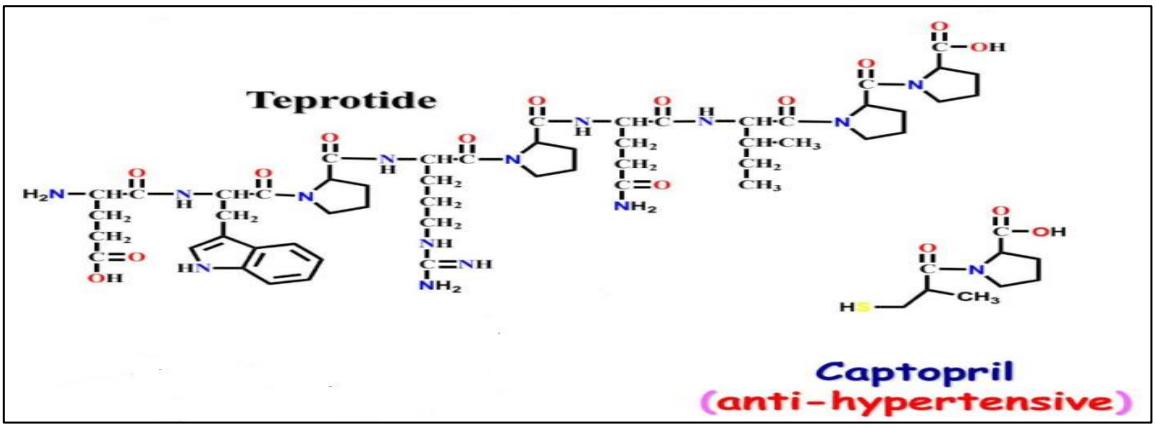


✓ **Venoms and toxins**

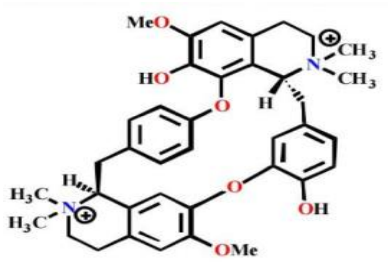
- **Teprotide**

  Nonapeptide isolated from a type of snake (not taken orally)

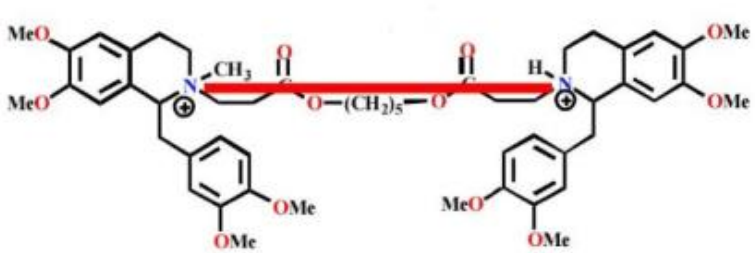
  Lead compound for ACEI



- **Tubocurarine**



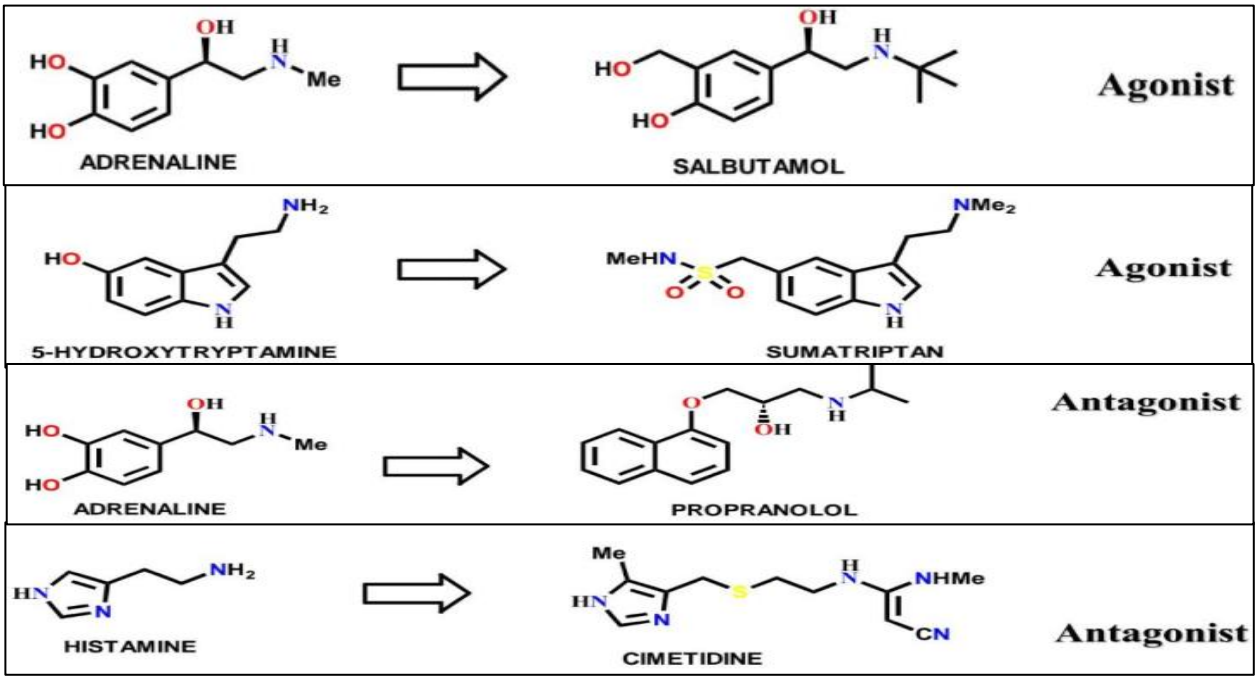
- **Atracurium**



Distance between two quaternary nitrogen in both compound is 11 A

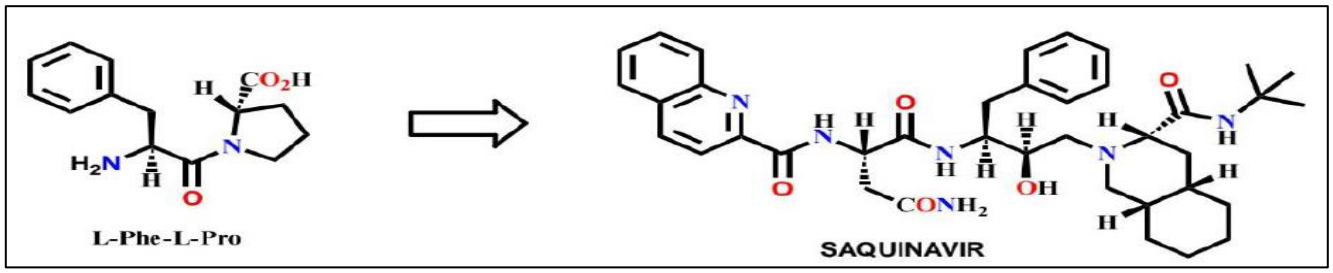
✓ Endogenous compounds

- Natural ligands for receptors



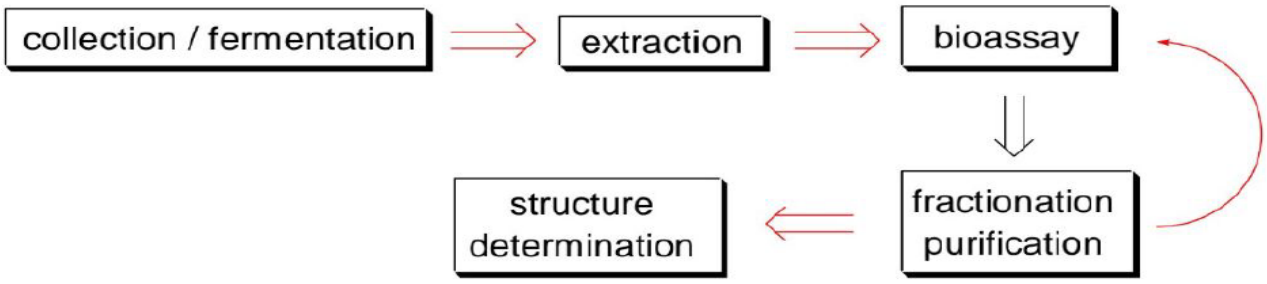
- Natural substrates for enzymes

Enkephalins ⟹ Enkephalinase inhibitors
Peptides ⟹ Protease inhibitors

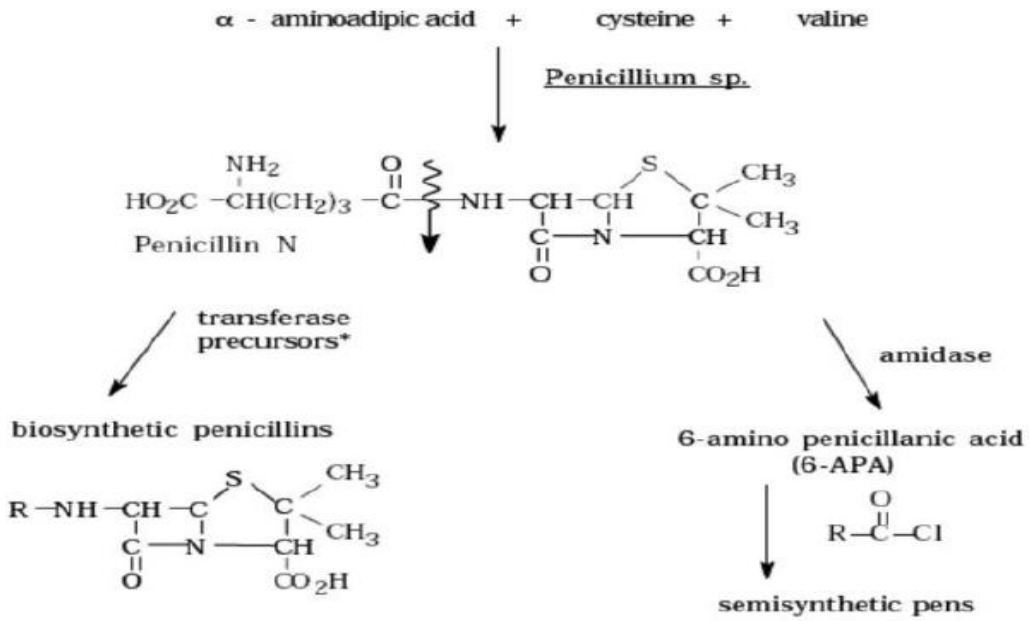


✓ Lead Discovery: Natural Products

- Terrestrial sources
  bacteria, fungi, plants, insects, reptiles, etc. importance of folk medicine
- Marine sources
  algae, bacteria, sponges, coral, fish, etc.

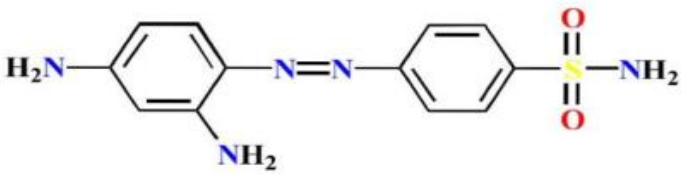


3. 4. Lead components from SemiSynthetic



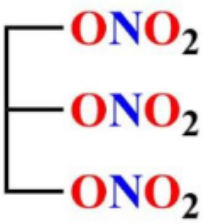
3. 5. Lead components from Synthetic World

- **Prontosil**



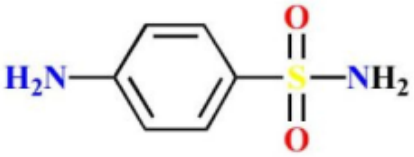
Azo dye has antibacterial activity in vivo
Considered as a prodrug

- **TNT**



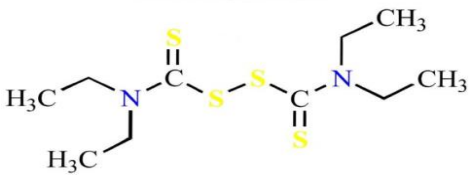
2,4,6-trinitrotoluene
Explosive material

- **sulfanilamide**



prototype for Sulfa drugs

- **Antabuse**

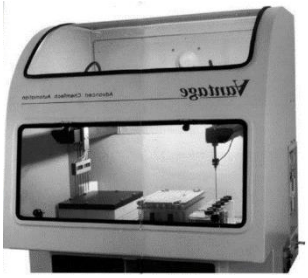


used to support the treatment of chronic alcoholism by inhibiting the enzyme aldehyde dehydrogenase

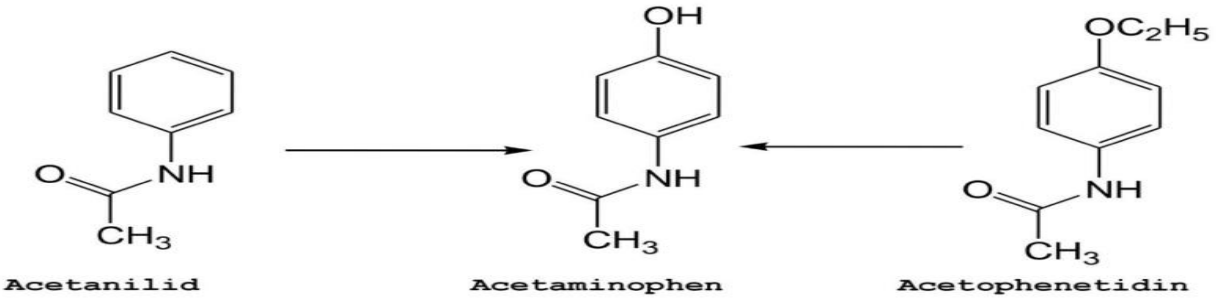
✓ **Organic synthesis**

✓ **Combinatorial synthesis**

Automated Synthetic machine
ex. (peptide synthesis)



3.6. Drug Metabolites as Leads for Drug Discovery

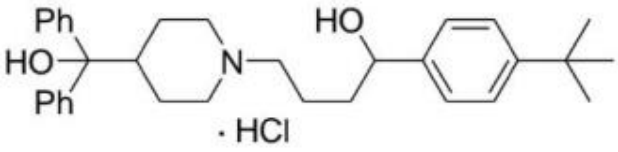


Acetanilid — Acetaminophen — Acetophenetidin

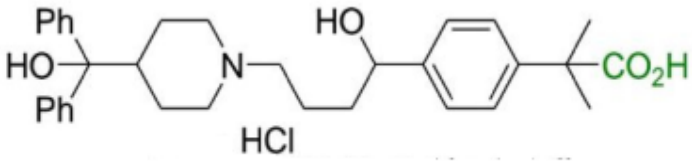
Drug Metabolism Studies

- **Terfenadine HCI** (Seldane):
  caused arrhythmia in some patients also taking certain antifungals.

- **Fexofenadine HCI** (Allegra):
  is a metabolite of terfenadine that is also a non-sedating antihistamine and can be metabolized in presence of antifungals.



terfenadine (Seldane)



Fexofenadine

3.7. Clinical Observations of Side Effects of Drugs

✓ **SOSA approach "selective optimization of side activities"**

Innumerable examples can be found in the literature, such as:

- The hypoglycemic effect of some antibacterial sulfonamides
- The antidepressant effect of isoniazid.
- The hypotensive effect of p-blocking agents.

✓ Clinical Observations



• Sildenafil (Viagra)

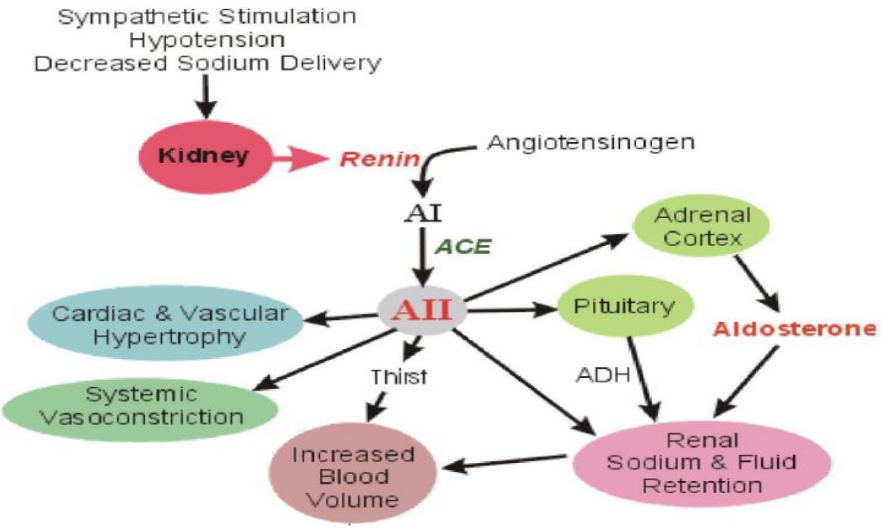
- Developed for angina and hypertension by Pfizer
- Lack of potency in Phase II trials sent the drug back to Phase I trials to determine highest tolerated dose
- Volunteers in second trial reported increased erectile function
- Became first treatment for erectile dysfunction

tadalafil (Cialis) and Vardenafil HCI (Levitra)

3.8. Rational Approaches

✓ The key information that permits rational approaches to drug design is knowledge of the etiology of a given disease or at least the biochemical processes that are disturbed.

✓ Examples of the rational approach are the discovery of the inhibitors of the angiotensin-converting enzyme and histamine H2-receptors antagonists.



3.9. Molecular Modification or Manipulation:

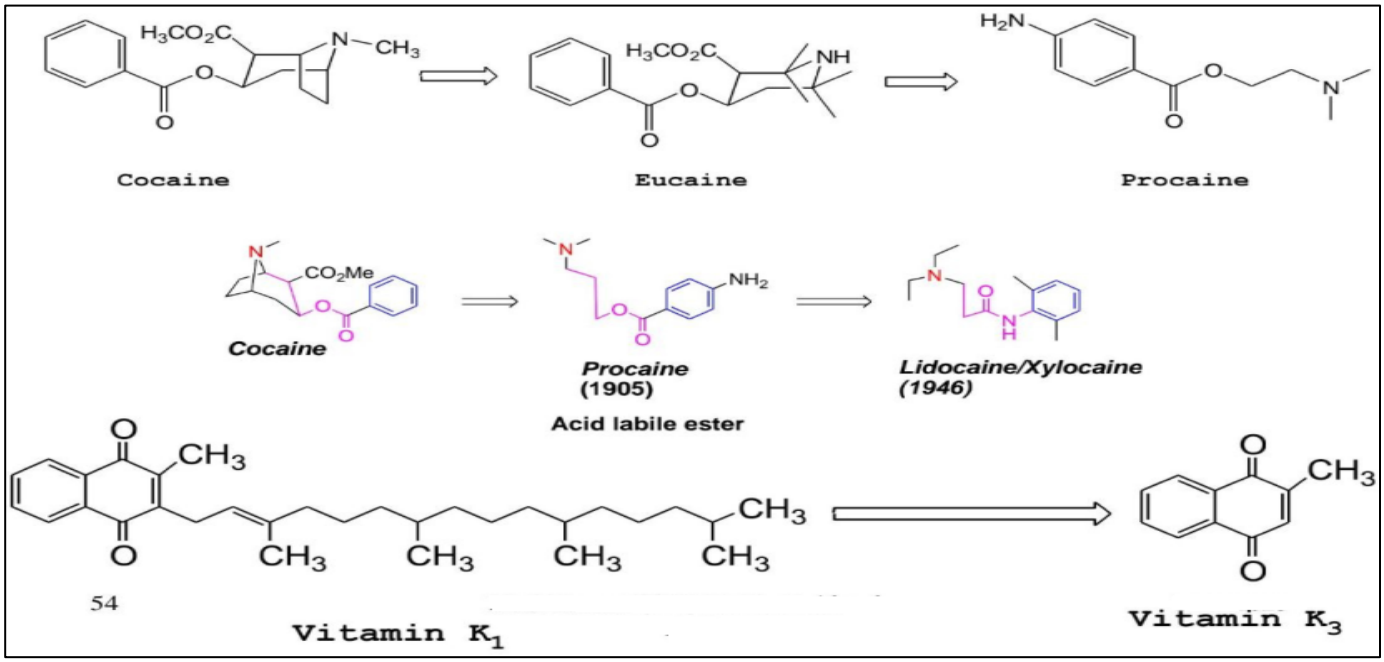
✓ To obtain drugs having more desirable properties than the prototype in potency, toxicity, specificity, duration of action, ease of application, stability and cost of production.

✓ Two main aspects can be differentiated in molecular manipulation

1. General Process
2. Special Process

1. General Process

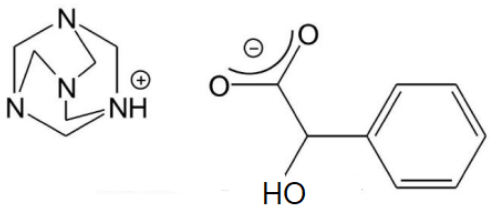
1.1. Disjunction or Molecular simplification



1.2. Conjunction or Molecular Association

• Three main types of association can be distinguished:

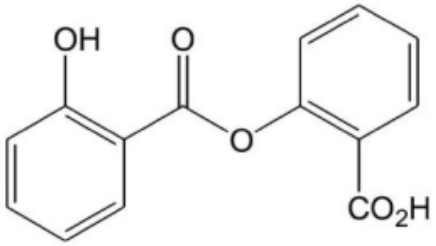
1. Molecular Addition:
association of different moieties through weak forces such as electrostatic attraction and hydrogen bonding.



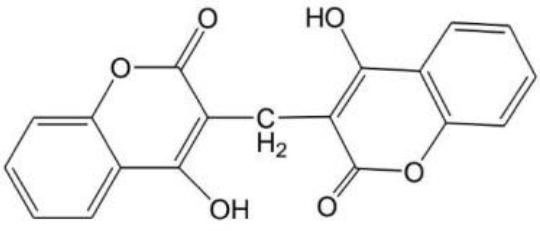
- example is methenamine mandelate

2. Molecular Replication:
association of identical moieties through covalent bond formation (Identical twin drug)

- e.g. Salicyl salicylate and dicumarol.

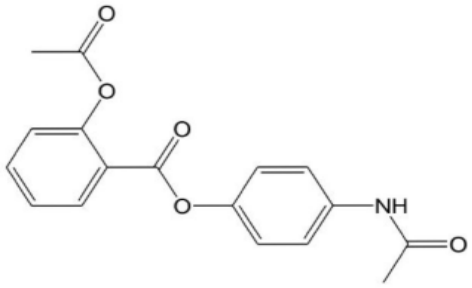


Salicyl salicylate



Dicumarol

3. Molecular Hybridization:
association of different or mixed moieties through covalent bond formation
(Non-identical twin drug)



- e.g. acetaminosalol (aspirin with acetaminophe).

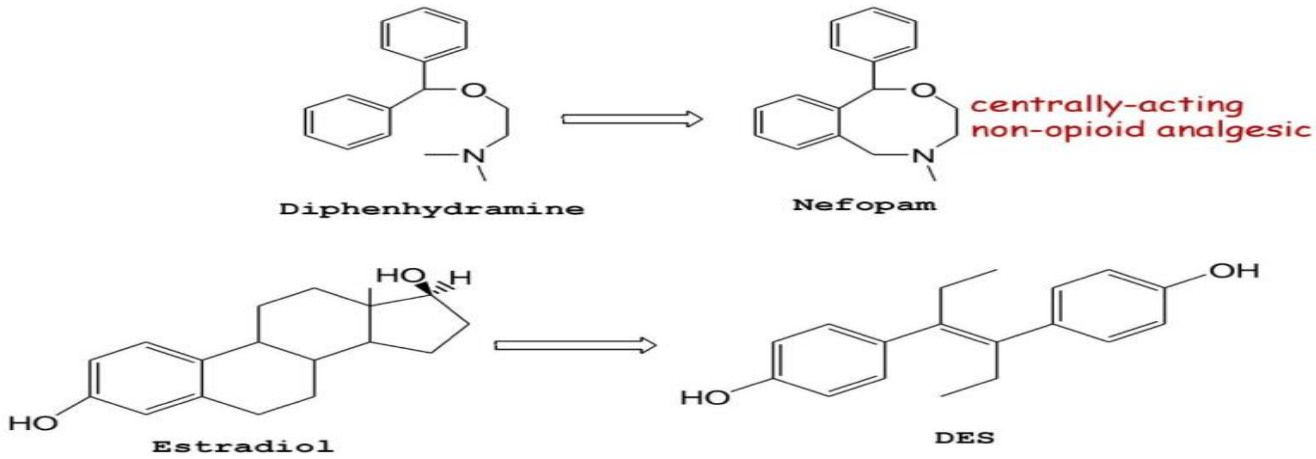
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2. Special Process

• Several special processes for molecular modification were grouped in two classes:

2.1. Alterations which increase or decrease the dimensions and flexibility of a molecule "Me-too compounds", this can be achieved through the following:

2.1.1. Ring Closure and Opening:



2.1.2 Ring Enlargement and Ring Contraction:

